

15CMO1

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Problem

Let $\mathbb{N} = \{1, 2, 3, \dots\}$ be the set of positive integers. Find all functions f , defined on $\mathbb{N}$ and taking values in $\mathbb{N}$, such that $(n-1)^2 < f(n)f(f(n)) < n^2 + n$ for every positive integer n .

Solution. We claim that $f(n) = n$ is the only function that works.

Claim 1 — If $f(a) = f(b)$ then $|a - b| \leq 1$.

Proof. Wolog say $b \geq a$. FTSOC assume $b \geq a + 2$.

$$\begin{aligned} \therefore (a-1)^2 &< f(a)f(f(a)) < a^2 + a \\ &< (a+1)^2 \\ &< (b-1)^2 < f(b)f(f(b)) = f(a)f(f(a)) \end{aligned}$$

Contradiction! □

We will proceed using induction.

Base case: $n = 1$

$$0 < f(1)f(f(1)) < 2$$

Since $f(1), f(f(1)) \in \mathbb{N}$, $\therefore f(1) = 1$.

Inductive step: Assume that $f(k) = k$ for all $k \in \{1, 2, \dots, n-1\}$. if $f(n) = c$ for some $c \in \{1, 2, \dots, n-1\}$ then by Claim 1, $c = n-1$.

$$\begin{aligned} \implies f(f(n)) &= n-1 \\ \therefore (n-1)^2 &< f(n)f(f(n)) = (n-1)^2 \end{aligned}$$

impossible!

If $f(n) = n$ then we are done!

If $f(n) > n$ then

$$nf(f(n)) < f(n)f(f(n)) < n^2 + n$$

$$\implies f(f(n)) < n + 1$$

If $f(f(n)) \leq n - 1 = f(n - 1)$ then due to Claim 1, $f(n) \leq (n - 1) + 1 = n$, contradiction!

If $f(f(n)) = n$ then

$$n^2 + n > f(n)f(f(n)) \geq (n + 1)n = n^2 + n$$

contradiction!

$\therefore f(n) = n$ is the only function that works.